

Module 1: Network + Network Fundamentals and building networks

OSI Model

Task 1:

7. Application

HTTP, browsers, email clients

6. Presentation

SSL/TLS encryption, JPEG, ASCII

5. Session

API session tokens, NetBIOS

4. Transport

TCP, UDP (segments)

3. Network

Router, IP addressing (packets)

2. Data link

Switch, MAC address (frames)

1. Physical

Cables, hubs, NIC (bits/signals)

Task 2:

Matching scenarios to OSI layers

- a) Encrypting data before sending: **Presentation Layer (6)**
- b) Routing packets across networks: **Network Layer (3)**
- c) Converting data to electrical signals: **Physical Layer (1)**
- d) Opening a web page in your browser: **Application Layer (7)**
- e) Detecting and correcting errors in frames: **Data Link Layer (2)**

Task 3:

OSI layers in a daily app (WhatsApp)

When you send a text/voice message on WhatsApp, several layers work together:

- **Application Layer (7):** WhatsApp itself, running over HTTPS, formats and initiates your message as a request the app understands.
- **Presentation Layer (6):** The message content is encrypted (WhatsApp uses end-to-end encryption) and compressed (for images/voice notes) so it can travel securely and efficiently.
- **Transport Layer (4):** TCP (or sometimes UDP for calls) breaks the message into segments, adds sequencing, and ensures reliable delivery to the recipient's device.
- **Network Layer (3):** IP addressing and routing get that data across the internet from your phone's network to the recipient's, potentially through several routers.
- **Physical Layer (1):** Your phone's Wi-Fi/mobile radio converts all this data into actual radio signals or electrical pulses to physically transmit it.

Task 4:

Website unreachable, but ping (by IP) works

Since ping succeeds, the **Network Layer (3)** and below (routing, IP connectivity, physical link) are all working fine — the path to the server exists and basic connectivity is confirmed.

The problem is most likely at the **Application Layer (7)** or **Transport Layer (4)**:

Application Layer (7): Most commonly this is a **DNS resolution failure** — DNS is technically an application-layer protocol, so if your browser can't resolve the domain name to an IP address, typing the URL fails even though the IP itself is reachable. It could also be that the web server (HTTP/HTTPS service) itself is down or misconfigured, even though the machine hosting it is alive.

Transport Layer (4): The specific port the website uses (e.g., port 80 or 443) might be blocked by a firewall or the web service isn't listening on it, so the TCP handshake for HTTP/HTTPS fails even though ICMP (ping) traffic gets through fine.